OUTCOMES-BASED EDUCATION (OBE) COURSE SYLLABUS IN
ITP831 – NETWORKING 2
Date: **January 2026**

I.	**School Year/Semester**	AY 2025-2026, 2nd Semester
II.	**University**	Vision: A globally-recognized green research University committed to excellent service to the nation. Mission: Foster excellence, holistic and inclusive education attuned to the demands of a diverse world towards the formation of productive and value-laden lives. Goal: Uphold the tradition of excellence in instruction, research, extension and production geared towards a strong academe-industry interface for societal development. Core Values: Respect, Integrity, Social Responsibility, Excellence, Commitment Institutional Outcomes: A. A *Professiona*l who is morally upright, socially responsible and globally employable B. A *Leader and Innovator* who inspires others and is committed to social transformation and national development C. An *Environmental Advocate* committed to research, extension and production initiatives
III.	**College/Campus**	Main Campus, Calatagan, Virac, Catanduanes
IV.	**Program/Degree**	Bachelor of Science in Information Technology (BSInfoTech)
V.	Program Outcomes	

A.	Articulate and discuss the latest developments in the specific field of practice.	D
B.	Effectively communicate orally and in writing using both English and Filipino.	I
C.	Work effectively and independently in multi-disciplinary and multi-cultural teams.	I
D.	Act in recognition of professional, social, and ethical responsibility.	P
E.	Preserve and promote “Filipino historical and cultural heritage”.	I
F.	Apply knowledge of computing science, and mathematics appropriate to the discipline.	P
G.	Understand best practices and standards and their applications	D
H.	Analyze complex problems and identify and define the computing requirements appropriate to its solution.	P
I.	Identify and analyze user needs and take them into account in the selection, creation, evaluation, and administration of computer-based systems.	D
J.	Design, implement, and evaluate computer-based systems, processes, components, or programs to meet desired needs and requirements under various constraints.	P
K.	Integrate IT-based solutions into the user environment effectively.	P
L.	Apply knowledge through the use of current techniques, skills, tools and practices necessary for the IT profession.	P

	M.	Function effectively as a member or leader of a development team recognizing the different roles within a team to accomplish a common goal.	I	
	N.	Assist in the creation of an effective IT project plan.	P	
	O.	Communicate effectively with the computing community and with society at large about complex computing activities through logical writing, presentations, and clear instructions.	I	
	P.	Analyze the local and global impact of computing information technology on individuals, organizations, and society.	P	
	Q.	Understand professional, ethical, legal, security and social issues and responsibilities in the utilization of information technology.	P	
	R.	Recognize the need for and engage in planning self-learning and improving performance as a foundation for continuing professional development.	I	
	S.	Participate in the generation of new knowledge or in research and development projects	I	
	T.	Support local, regional, and national development plans along education, environment, socio economic, health, gender and development, science and technology.	I	

I – Introduced **P-Practiced** **D-Demonstrated**

VI. Course Code/Title	ITP831/ Networking 2
VII. Course Description	This course provides an advanced knowledge of networking as well as related equipment and terminologies. The course will cover Local Area Network, Wide Area Network, managing enterprise level networks using Active Directory and remote access. Advanced network management and environment customization techniques will be explored, including creating user/ groups, managing file permissions, configure server roles, using group policies to configure and secure the network, routine system maintenance and troubleshooting.
VIII. Course Credit	3 units (2 units Lecture/ 1 unit Laboratory)
IX. Prerequisite	ITP422/ Networking I
X. Contact Hours	5 contact hours per week (2 hours lecture, 3 hours laboratory)
XI. Course Outcomes	At the end of the course, the students shall be able to:

COs	Description	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T
CO1	Apply basic concepts of System Administration and Security	3			3		2	2	1	1	2	3	3		2		1	3			
CO2	Demonstrate knowledge and comprehension about the internal operation of communications between computers	3			3		2	2	1	1	2	3	3		2		1	3			
CO3	Evaluate and select hardware and software production platforms for	3			3		2	2	1	1	2	3	3		2		1	3			

		executing applications and computer services																				
	CO4	Design, deploy, maintain and manage computer networks	3			3		2	2	1	1	2	3	3		2		1	3			

XII. Course Outline/Learning Plan

POs	COs	Desired Learning Outcomes	Course Content/ Subject Matter	Textbooks/ References	Teaching and Learning Activities *(TLAs)*	Assessment Tasks	Instructional Resources/ Materials	Time Table (Hours)
A D F G H I J K L N Q	CO1 CO2 CO3 CO4	At the end of the lesson the students shall be able to: 1. Explain and Differentiate Core Networking Devices and Concepts 2. Summarize Key Principles of Cloud Computing in Networking Contexts 3. Compare and Apply Knowledge of IPv6 and IPv5 Addressing	A. Introduction and Review of Networking 1. Review of Switch, Router, VLAN 2. Review of Cloud Computing 3. Review of IPV6 and IPV5	Solomon & Kim.(2022), Fundamentals of Communications and Networking Third Edition.Jones and Bartlett Learning Lowe, Doug(2021), *Networking All-in-One 8th Edition*. John Wiley & Sons, Inc. Sadiku and Akujuobi (2022). *Fundamentals of Computer Networks*. Springer Nature Switzerland. e-ISBN 978-3-031-09417-0	• Lecture and classroom discussion • Brainstorming • Interactive discussion • Guided Hands-on Activities • Flexible learning activities (online and/or offline modality) in times of class disruptions	▪ Graded Recitation ▪ Quizzes ▪ Homework ▪ Seatwork	▪ Handouts ▪ Ppt Presentation ▪ Projector/TV ▪ PC/ Laptop/ Cellphones ▪ Internet Connectivity	15
A D L Q	CO1 CO2 CO3 CO4	1. Design and Implement OSPF Hierarchical Networks	B. Advance Routing Protocol 1. OSPF (Open Short Path First	Solomon & Kim.(2022) Lowe, Doug(2021)	• Lecture and classroom discussion • Brainstorming	▪ Graded Recitation ▪ Quizzes ▪ Homework ▪ Seatwork	▪ Handouts ▪ Ppt Presentation ▪ Projector/TV	15

		2. Configure and Manage BGP for Inter- and Intra-Domain Routing 3. Optimize EIGRP for Performance and Reliability	A. Area design and hierarchy B. LSA Type and Database C. OSPF Configuration and Optimization 2. BGP (Boarder Gateway Protocol) A. eBGP vs iBGP B. Path selection and Route policies C. BGP communities and route filtering 3. EIGRP Advance feature A. Metric calculation B. Load balancing and Convergence optimization	Sadiku and Akujuobi (2022).	• Interactive discussion • Guided Hands-on Activities • Flexible learning activities (online and/or offline modality) in times of class disruptions		▪ PC/ Laptop/ Cellphones ▪ Internet Connectivity	
A D F G H I J K L N	CO1 CO2 CO3 CO4	1. Apply Hierarchical Network Design Principles 2. Implement Network Convergence and High Availability Solutions	C. Network Design Pattern 1. Hierarchical Network Design A. Core, Distribution and Access layer B. Scalability and redundancy planning 2. Network convergence and High Availability A. Redundant path protocol B. First Hop Redundancy	Solomon & Kim.(2022) Lowe, Doug(2021) Sadiku and Akujuobi (2022).	• Lecture and classroom discussion • Brainstorming • Interactive discussion • Guided Hands-on Activities • Flexible learning activities (online and/or offline modality) in times of class disruptions	▪ Graded Recitation ▪ Quizzes ▪ Homework ▪ Seatwork	▪ Handouts ▪ Ppt Presentation ▪ Projector/TV ▪ PC/ Laptop/ Cellphones ▪ Internet Connectivity	12.5

			Protocol (HSRP, VRRP, GLBP)					
MID-TERM EXAMINATION								**2.5**
A D F G H I J K L N	CO1 CO2 CO3	1. Design and Configure Advanced VLAN and Spanning Tree Solutions 2. Enhance Network Security through Layer 2 Protection Mechanisms	D. Advance Switch and VLAN 1. Advance VLAN Technologies A. VLAN Trunking and Inter-VLAN Routing a. 802.1Q tagging and Native VLANs b. Router-on-a-stick vs Layer 3 VLANs B. Advance Spanning tree Protocol a. RSTP, MST configuration and optimization b. PortFast, BPDU Guard and Root Guard 2. Layer Security and Optimization A. Port Security and 802.1X a. MAC Address timing & Violation action b. Network Access Control (NAC) B. VLAN Security best Practices	Axel, Ross(2022). Computer Networking Beginners Guide: An Introduction on Wireless Technology and Systems Security to Pass CCNA Exam, With a Hint of Linux Programming and Command Line. ISBN 1001921680, 1001921684, 1001921688	• Lecture and classroom discussion • Brainstorming • Interactive discussion • Guided Hands-on Activities • Flexible learning activities (online and/or offline modality) in times of class disruptions	▪ Graded Recitation ▪ Quizzes ▪ Homework ▪ Seatwork	▪ Handouts ▪ Ppt Presentation ▪ Projector/TV ▪ PC/ Laptop/ Cellphones ▪ Internet Connectivity	25

			a. Private VLAN & VLAN Hopping prevention b. Dynamic VLAN assignment					
A D L Q	CO1 CO2 CO3 CO4	1. Implement QoS Techniques for Traffic Prioritization and Control 2. Optimize Network Performance with Advanced Traffic Management	E. Quality of Service(QoS) and Traffic Management 1. QoS Fundamental and Implementation A. Traffic Classification and Marking a. DSCP and CoS Marking (Differentiated Services Code Point, Class of Service) b. Traffic policing and shaping B. Queuing Mechanism c. Priority queuing, weighted fair queuing d. Class-based weighted fair queuing (CBWFQ) 2. Advance QoS and Traffic Management A. Traffic Engineering with MPLS a. Label Switching and LSP Establishment b. MPLS VPN Fundamentals	Lowe, Doug(2021) Sadiku and Akujuobi (2022).	• Lecture and classroom discussion • Brainstorming • Interactive discussion • Guided Hands-on Activities • Flexible learning activities (online and/or offline modality) in times of class disruptions	▪ Graded Recitation ▪ Quizzes ▪ Homework ▪ Seatwork	▪ Handouts ▪ Ppt Presentation ▪ Projector/TV ▪ PC/ Laptop/ Cellphones ▪ Internet Connectivity	10

			B. Congestion Management c. Random Early Detection (RED) d. Traffic load balancing strategies					
A D F G H I J K L N	CO1 CO2 CO3 CO4	1. Implement and Manage Network Security Infrastructure 2. Deploy and Secure VPN Communication Channels 3. Apply Advanced Security Protocols for Access Control and Data Protection	F. Network Security and Advance Protocol 1. Network Security Infrastructure A. Firewall Technology a. Stateful vs Stateless firewall b. Next Generation Firewall (NGFW) c. DMZ Design and Implementation B. Intrusion Detection and Prevention system d. Network-based vs Host-based IDS/IPS e. Signature-based vs Anomaly-based detection 2. VPN Technologies and Secure Communication A. Site-to-Site VPN	Lowe, Doug(2021) Sadiku and Akujuobi (2022).	• Lecture and classroom discussion • Brainstorming • Interactive discussion • Guided Hands-on Activities • Flexible learning activities (online and/or offline modality) in times of class disruptions	▪ Graded Recitation ▪ Quizzes ▪ Homework ▪ Seatwork	▪ Handouts ▪ Ppt Presentation ▪ Projector/TV ▪ PC/ Laptop/ Cellphones ▪ Internet Connectivity	7.5

			a. IPSec protocol and Configuration b. GRE Tunnel and Encryption B. Remote Access VPN c. SSL/TLS VPN Implementation d. Certificate management and PKI 3. Advance Security Protocol A. Network Access Control (802.1X) e. EAP method and RADIUS Integration f. Wireless security (WP3, Enterprise mode) B. DNS Security and DNSSEC g. DNS over HTTPS (DoH) and DNS over TLS (DoT) h. DNS filtering and threat protection					
FINAL EXAMINATION								**2.5**
XII. Suggested Readings and References			1. Bajd, T., et al. (2019), *Robotics*, Springer International Publishing 2. Lowe, D. (2018), Networking All-in-One for Dummies, For Dummies					

	3. Joshi, A., et al. (2018), *Emerging Trends in Expert Applications and Security: Proceedings of ICETEAS 2018*, Springer Singapore 4. Mathew, Jack. (2021). *Networking for Beginners_ Simple Guide to Study Basic_Advanced Computer Network, Hardware, Wireless, and Cabling. LTE, Internet, and Cyber Security* . 5. West, J. (2019), *Network+ Guide to Networks Eight Edition*. Cengage
XIII. Course Requirements	1. Interactive discussion 2. Group or Individual activity/group report 3. Quizzes/Seatwork 4. Homework/Hands-on Activities 5. Completed Project or Case study to be submitted on or before Final Examination 6. Mid-term and final examination
XIV. Course Policies	1. Students with reasonable absences from the class will be given consideration for the missed quizzes/ examinations upon presentation of a duly signed excuse letter. 2. Students who incurred absences of more than 20% of the required number of hours shall be dropped from the rolls. 3. Submission of outputs shall be on or before the due date agreed upon by the students and the faculty. Late submission will earn a grade of 3.0. 4. Use of cellphones, earphones, and other gadgets that are not necessary for the instruction is strictly prohibited. 5. Cheating during examination is strictly prohibited; students caught doing such act shall be dealt with as provided in the Student Handbook. 6. In the event of class disruptions, the Guidelines for Flexible Learning Modality will take effect with the university's approved provisions in shifting classes.
XV. Grading System (Per Board Resolution No.3,4 & 5 s. 2016)	Midterm/Final Examination 30% Written Output 30% Performance 40% TOTAL 100% Distribution of Weight for the Overall/ Final Grade: Midterm Grade 50% Tentative 50%
XVI. Consultation Time/Venue	MTh 1:00pm – 4:00pm – CICT FACULTY OFFICE

Prepared by:

JOY V. SANTELICES, MSIT
Assoc. Prof. III
Date: _____

Reviewed by:

ASTER VIVIEN C. VARGAS, MSIT
Department Chairperson, Computing Programs
Date:_____

Approved:

MARIA CONCEPCION S. VERA, DIT
Dean, CICT
Date: _____